

Chapter 6 — Capabilities, Providers, Actions, Transactions, and the Reality Boundary

Purpose and scope

Defines abilities, implementations, governed execution, commit barriers, rollback, compensation, and separation of simulated from real effects.

Source authority

Book I: Book I Decisions 10, 15, 16, 20, 23, 24, 35, 36, 44, and 50

Book II: Book II Chapters 15, 16, 17, 22, 27, and 28

Book III: Book III Chapters 03, 05, 06, and 07

Normative semantic rules

- A Capability MUST define an implementation-independent ability; Provider and Adapter records MUST identify implementations without redefining that ability.
- Every real Action MUST cross an explicit Commit Barrier with applicable Authority, Permission, verification, and evidence.
- Simulation, Digital Twin, and Shadow Execution MUST remain incapable of ungoverned real effects.
- Rollback MUST be used only for truthful reversal; Compensation MUST name a new remedial Action.

Canonical terms

HAL-TERM-0062 — Capability

An abstract ability defined by outcomes, inputs, outputs, constraints, required authority and permission classes, risks, side effects, and evaluation criteria.

Distinction: Capability does not identify an implementation and does not grant authority to use the ability.

Type/category/status: Implementation-independent contract / Capability / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36

Book II: Book II Chapter 15

Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0062` when it uses **Capability** with this exact governed meaning: An abstract ability defined by outcomes, inputs, outputs, constraints, required authority and permission classes, risks, side effects, and evaluation criteria.

Counterexample: A dependent artifact uses **Capability** in a way that violates its required distinction: Capability does not identify an implementation and does not grant authority to use the ability.

Relationships: HAL-REL-0021, HAL-REL-0022

Lifecycle: None registered

Constraint: Capability does not identify an implementation and does not grant authority to use the ability.

HAL-TERM-0063 — Capability Contract

The versioned specification of a Capability's semantic inputs, outputs, preconditions, effects, risks, authority requirements, evidence, and compatibility.

Distinction: It is not a provider-specific API contract.

Type/category/status: Contract record / Capability / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36

Book II: Book II Chapter 15

Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0063` when it uses **Capability Contract** with this exact governed meaning: The versioned specification of a Capability's semantic inputs, outputs, preconditions, effects, risks, authority requirements, evidence, and compatibility.

Counterexample: A dependent artifact uses ****Capability Contract**** in a way that violates its required distinction: It is not a provider-specific API contract.

Relationships: HAL-REL-0021

Lifecycle: None registered

Constraint: It is not a provider-specific API contract.

HAL-TERM-0064 — Provider

A component, service, model, person, device, or external system that can fulfill a Capability under a declared contract and trust context.

Distinction: Being able to perform work does not authorize the Provider to perform it.

Type/category/status: Implementation role / Capability / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36

Book II: Book II Chapter 15

Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0064` when it uses ****Provider**** with this exact governed meaning: A component, service, model, person, device, or external system that can fulfill a Capability under a declared contract and trust context.

Counterexample: A dependent artifact uses ****Provider**** in a way that violates its required distinction: Being able to perform work does not authorize the Provider to perform it.

Relationships: HAL-REL-0022, HAL-REL-0023

Lifecycle: None registered

Constraint: Being able to perform work does not authorize the Provider to perform it.

HAL-TERM-0065 — Adapter

A component that translates between a Capability Contract and a provider-specific interface while preserving authority, semantics, evidence, and failure behavior.

Distinction: An Adapter must not smuggle provider semantics into the canonical Capability definition.

Type/category/status: Boundary component role / Capability / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36

Book II: Book II Chapter 15

Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0065` when it uses ****Adapter**** with this exact governed meaning: A component that translates between a Capability Contract and a provider-specific interface while preserving authority, semantics, evidence, and failure behavior.

Counterexample: A dependent artifact uses ****Adapter**** in a way that violates its required distinction: An Adapter must not smuggle provider semantics into the canonical Capability definition.

Relationships: HAL-REL-0023

Lifecycle: None registered

Constraint: An Adapter must not smuggle provider semantics into the canonical Capability definition.

HAL-TERM-0066 — Capability Registry

The governed catalog of Capabilities, versions, Providers, constraints, authority classes, health, and selection metadata.

Distinction: Registration does not establish permission for use or trustworthiness in every domain.

Type/category/status: Authoritative registry / Capability / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36

Book II: Book II Chapter 15

Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0066` when it uses ****Capability Registry**** with this exact governed meaning: The governed catalog of Capabilities, versions, Providers, constraints, authority classes, health, and selection metadata.

Counterexample: A dependent artifact uses ****Capability Registry**** in a way that violates its required distinction: Registration does not establish permission for use or trustworthiness in every domain.

Relationships: None registered

Lifecycle: None registered

Constraint: Registration does not establish permission for use or trustworthiness in every domain.

HAL-TERM-0067 — Action

A governed attempt to produce an effect in authoritative state or the external world.

Distinction: A read-only Query is not an Action; an Action is not proof of successful effect.

Type/category/status: State-changing attempt / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0067` when it uses ****Action**** with this exact governed meaning: A governed attempt to produce an effect in authoritative state or the external world.

Counterexample: A dependent artifact uses ****Action**** in a way that violates its required distinction: A read-only Query is not an Action; an Action is not proof of successful effect.

Relationships: HAL-REL-0024

Lifecycle: None registered

Constraint: A read-only Query is not an Action; an Action is not proof of successful effect.

HAL-TERM-0068 — Attempt

One attributable execution effort for a Task, Action, or verification step with its own timing, context, result, and evidence.

Distinction: A retry is a new Attempt even when it shares an idempotency key.

Type/category/status: Execution instance / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0068` when it uses ****Attempt**** with this exact governed meaning: One attributable execution effort for a Task, Action, or verification step with its own timing, context, result, and evidence.

Counterexample: A dependent artifact uses ****Attempt**** in a way that violates its required distinction: A retry is a new Attempt even when it shares an idempotency key.

Relationships: None registered

Lifecycle: None registered

Constraint: A retry is a new Attempt even when it shares an idempotency key.

HAL-TERM-0069 — Transaction

The durable coordination object for one or more Actions, including authorization, prepare, commit, result, evidence, rollback, and compensation states.

Distinction: It is broader than a database transaction and must not imply atomic reversibility of external effects.

Type/category/status: Governed action lifecycle / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0069` when it uses ****Transaction**** with this exact governed meaning: The durable coordination object for one or more Actions, including authorization, prepare, commit, result, evidence, rollback, and compensation states.

Counterexample: A dependent artifact uses ****Transaction**** in a way that violates its required distinction: It is broader than a database transaction and must not imply atomic reversibility of external effects.

Relationships: HAL-REL-0024

Lifecycle: HAL-TRANS-0008, HAL-TRANS-0009, HAL-TRANS-0010, HAL-TRANS-0011, HAL-TRANS-0012, HAL-TRANS-0013

Constraint: It is broader than a database transaction and must not imply atomic reversibility of external effects.

HAL-TERM-0070 — Commit Barrier

The explicit governed point after which a proposed change may create authoritative or real-world effects that cannot be treated as merely simulated or prepared.

Distinction: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.

Type/category/status: Irreversibility gate / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0070` when it uses ****Commit Barrier**** with this exact governed meaning: The explicit governed point after which a proposed change may create authoritative or real-world effects that cannot be treated as merely simulated or prepared.

Counterexample: A dependent artifact uses ****Commit Barrier**** in a way that violates its required distinction: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.

Relationships: None registered

Lifecycle: None registered

Constraint: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.

HAL-TERM-0071 — Rollback

A controlled restoration of a prior recoverable state when the relevant effects are truthfully reversible.

Distinction: Rollback must not claim to erase external or human effects that already occurred.

Type/category/status: Reversal operation / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0071` when it uses ****Rollback**** with this exact governed meaning: A controlled restoration of a prior recoverable state when the relevant effects are truthfully reversible.

Counterexample: A dependent artifact uses ****Rollback**** in a way that violates its required distinction: Rollback must not claim to erase external or human effects that already occurred.

Relationships: None registered

Lifecycle: None registered

Constraint: Rollback must not claim to erase external or human effects that already occurred.

HAL-TERM-0072 — Compensation

A new governed Action that mitigates, offsets, or repairs effects that cannot truthfully be undone.

Distinction: Compensation is not Rollback and may require independent authority and evidence.

Type/category/status: Remedial operation / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50

Book II: Book II Chapter 16

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0072` when it uses ****Compensation**** with this exact governed meaning: A new governed Action that mitigates, offsets, or repairs effects that cannot truthfully be undone.

Counterexample: A dependent artifact uses ****Compensation**** in a way that violates its required distinction: Compensation is not Rollback and may require independent authority and evidence.

Relationships: None registered

Lifecycle: None registered

Constraint: Compensation is not Rollback and may require independent authority and evidence.

HAL-TERM-0073 — Reality Boundary

The explicit separation among simulation, digital twin, shadow, test, canary, controlled-reality, production, recovery, and emergency contexts.

Distinction: Non-reality authority, data, or effects must not leak across this boundary into reality.

Type/category/status: Governed environment boundary / Action / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50

Book II: Book II Chapter 17

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0073` when it uses ****Reality Boundary**** with this exact governed meaning: The explicit separation among simulation, digital twin, shadow, test, canary, controlled-reality, production, recovery, and emergency contexts.

Counterexample: A dependent artifact uses ****Reality Boundary**** in a way that violates its required distinction: Non-reality

authority, data, or effects must not leak across this boundary into reality.

Relationships: HAL-REL-0025

Lifecycle: None registered

Constraint: Non-reality authority, data, or effects must not leak across this boundary into reality.

HAL-TERM-0074 — Simulation

An execution environment whose effects are confined to modeled or synthetic state and cannot directly alter production or external reality.

Distinction: High apparent fidelity does not make a Simulation production.

Type/category/status: Non-reality environment / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50

Book II: Book II Chapter 17

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0074` when it uses **Simulation** with this exact governed meaning: An execution environment whose effects are confined to modeled or synthetic state and cannot directly alter production or external reality.

Counterexample: A dependent artifact uses **Simulation** in a way that violates its required distinction: High apparent fidelity does not make a Simulation production.

Relationships: None registered

Lifecycle: None registered

Constraint: High apparent fidelity does not make a Simulation production.

HAL-TERM-0075 — Digital Twin

A governed model of selected real entities, relationships, state, and dynamics used to evaluate behavior without treating modeled effects as real effects.

Distinction: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.

Type/category/status: Modeled counterpart / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50

Book II: Book II Chapter 17

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0075` when it uses **Digital Twin** with this exact governed meaning: A governed model of selected real entities, relationships, state, and dynamics used to evaluate behavior without treating modeled effects as real effects.

Counterexample: A dependent artifact uses **Digital Twin** in a way that violates its required distinction: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.

Relationships: None registered

Lifecycle: None registered

Constraint: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.

HAL-TERM-0076 — Shadow Execution

Execution using live or representative inputs while preventing proposed outputs from producing authoritative or external effects.

Distinction: Shadow results do not authorize promotion without the required review and certification.

Type/category/status: Non-committing execution mode / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50

Book II: Book II Chapter 17

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0076` when it uses **Shadow Execution** with this exact governed meaning: Execution using live or representative inputs while preventing proposed outputs from producing authoritative or external effects.

Counterexample: A dependent artifact uses **Shadow Execution** in a way that violates its required distinction: Shadow results do not authorize promotion without the required review and certification.

Relationships: None registered

Lifecycle: None registered

Constraint: Shadow results do not authorize promotion without the required review and certification.

HAL-TERM-0077 — Canary

A deliberately constrained real execution stage used to accumulate evidence before broader adoption.

Distinction: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.

Type/category/status: Limited reality stage / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50

Book II: Book II Chapter 17

Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0077` when it uses **Canary** with this exact governed meaning: A deliberately constrained real execution stage used to accumulate evidence before broader adoption.

Counterexample: A dependent artifact uses **Canary** in a way that violates its required distinction: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.

Relationships: HAL-REL-0025

Lifecycle: None registered

Constraint: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.

Relationship and lifecycle rules

Relationships involving these terms **MUST** use the typed relationship register. Lifecycle-bearing concepts **MUST** use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.